Statistics Interview Question Bank

Section 1: Descriptive Statistics & Data Understanding

1. Customer Spending Analysis

An e-commerce company reports that the **average customer spending increased from ₹2,100 to ₹2,450** after introducing a loyalty programme. However, you notice that a small group of premium customers made unusually large purchases during the same period.

Questions:

- Would you conclude that the loyalty programme was successful based only on the average spending?
- Which descriptive statistics would you analyse before presenting your findings?
- How would you explain your conclusion to a business stakeholder?

Skills Assessed: Mean, Median, Outliers, Business Interpretation

2. Manufacturing Quality Control

A manufacturing plant produces metal rods with a target length of **100 cm**. During quality inspection, you find that today's production has the same average length as yesterday's, but customer complaints have increased.

Questions:

- What statistical measure would help explain this situation?
- Why isn't the mean sufficient in this case?
- What additional analysis would you perform?

Skills Assessed: Standard Deviation, Variability, Quality Analysis

3. Call Centre Performance

A customer support team claims that the **average call handling time has reduced by 15%** after introducing an AI-assisted support tool.

After reviewing the data, you notice that while experienced agents have become faster, new agents are taking much longer than before.

Questions:

- Is reporting only the average call handling time appropriate?
- Which descriptive statistics or visualisations would provide a clearer picture?

- How would you present this analysis to management?

Skills Assessed: Mean vs Distribution, Box Plots, Data Interpretation

4. Salary Analysis

A company's HR dashboard reports that the **average annual salary is ₹11 lakh**. However, a few executives earn significantly higher salaries than the rest of the workforce.

Questions:

- Which measure of central tendency would best represent a typical employee's salary?
- Why might relying only on the mean be misleading?
- In which business situations would the mean still be useful?

Skills Assessed: Mean, Median, Skewed Data

5. Sales Dashboard Validation

A Power BI dashboard shows that **monthly sales have increased by 18%** compared to the previous month. Before presenting the dashboard to senior leadership, you discover that duplicate invoices were accidentally included during data refresh.

Questions:

- Which descriptive statistics would help identify this issue?
- What validation checks should every data analyst perform before publishing dashboards?
- How could this error impact business decisions?

Skills Assessed: Data Validation, Summary Statistics, Data Quality

6. Website Performance Monitoring

A company's website receives **50,000 daily visitors**. One day, traffic suddenly increases to **1,80,000 visitors**.

Questions:

- Would you immediately treat this as a positive business outcome?
- How would you determine whether this observation is an outlier or part of a normal trend?
- What additional information would you investigate?

Skills Assessed: Outlier Detection, Trend Analysis, Business Reasoning

7. Healthcare Analytics

A hospital analyses patient waiting times across different departments. The Emergency Department has the same average waiting time as the General Consultation Department, but patient complaints are significantly higher.

Questions:

- What descriptive statistics might explain this difference?
- Which visualisation would best communicate the variation in waiting times?
- How would your findings help hospital management?

Skills Assessed: Variability, Percentiles, Box Plot Interpretation

8. Product Rating Analysis

A newly launched product has an average customer rating of **4.4 out of 5**. However, most reviews are either **5 stars or 1 star**, with very few moderate ratings.

Questions:

- What does this distribution suggest about customer satisfaction?
- Which descriptive statistics would you analyse beyond the average rating?
- How might this influence business decisions?

Skills Assessed: Distribution Shape, Variance, Customer Analytics

9. AI-Assisted Resume Screening

An AI-powered recruitment system assigns candidates an average suitability score of **82/100**. During an audit, you notice that scores vary significantly across different job roles.

Questions:

- Is reporting a single average score sufficient?
- Which descriptive statistics would better represent the data?
- How would you communicate these findings to the HR team?

Skills Assessed: Group-wise Analysis, Spread, Business Communication

10. Retail Inventory Analysis

A retail chain reports that the **average inventory level across stores is 3,500 units**. However, some stores experience frequent stockouts while others have excess inventory.

Questions:

- Why might the average inventory level be misleading?
- Which additional statistics would you calculate?

- How could your analysis support better inventory planning?

Skills Assessed: Measures of Dispersion, Business Analytics, Decision Making

Section 2: Probability

11. Online Payment Success Rate

An online payment platform reports a **98.7% transaction success rate**. A customer experiences three failed transactions in succession and claims the platform is unreliable.

Questions:

- Does this necessarily indicate a system issue?
- Which probability concepts would help explain the situation?
- What additional data would you analyse before drawing conclusions?

Skills Assessed: Probability, Independent Events, Business Interpretation

12. Manufacturing Defects

A factory produces electronic components with a historical defect rate of **1.5%**. During quality inspection, 8 defective units are found in a sample of 150 components.

Questions:

- Is this unusual based on the historical defect rate?
- Which probability distribution would you use to analyse this scenario?
- What factors could explain the difference?

Skills Assessed: Binomial Probability, Statistical Reasoning

13. Insurance Risk Analysis

An insurance company uses historical claims data to estimate accident probabilities. A customer argues that because they haven't had an accident in the past five years, they're unlikely to have one in the future.

Questions:

- Is this reasoning statistically valid?
- Which probability concepts are relevant?
- How would you explain your answer to the customer?

Skills Assessed: Independence, Conditional Probability

14. Email Spam Detection

An email filtering system correctly classifies 97% of emails. However, spam emails represent only 2% of all incoming emails.

Questions:

- Why can overall accuracy be misleading?
- What additional probability-based metrics should be considered?
- How does class imbalance affect interpretation?

Skills Assessed: Base Rate, Conditional Probability, Model Evaluation

15. AI Recommendation System

A streaming platform recommends movies based on viewing history. Users who watch science fiction movies also frequently watch documentaries.

Questions:

- Does this imply one preference causes the other?
- Which probability concepts would help analyse this relationship?
- What additional data would strengthen your conclusion?

Skills Assessed: Joint Probability, Association vs Causation

Section 3: Sampling & Data Collection

16. Customer Satisfaction Survey

A retail chain surveys **400 customers** from one metropolitan city and concludes that customers nationwide are highly satisfied.

Questions:

- What concerns do you have about the sampling method?
- How might sampling bias affect the results?
- How would you improve the study?

Skills Assessed: Sampling Bias, Representativeness

17. Healthcare Research

Researchers want to estimate the average recovery time for patients across India but collect data from only one hospital.

Questions:

- Is this sample representative?
- What sampling strategy would you recommend?
- Why is representativeness more important than sample size alone?

Skills Assessed: Sampling Methods

18. Product Feedback

A software company receives feedback only from users who voluntarily submit reviews.

Questions:

- What type of sampling bias might exist?
- How could this influence product decisions?
- What additional data would you collect?

Skills Assessed: Self-selection Bias

19. AI Model Training

A fraud detection model is trained using transaction data collected only during weekdays.

Questions:

- What sampling issues could arise?
- How might this affect model performance?
- How would you improve the dataset?

Skills Assessed: Sampling Bias, Data Representativeness

20. Employee Engagement

HR surveys only employees who attended a leadership workshop and concludes that employee engagement has improved across the organisation.

Questions:

- Why might this conclusion be misleading?
- What sampling strategy would produce more reliable insights?

Skills Assessed: Selection Bias

Section 4: Probability Distributions & Central Limit Theorem

21. Delivery Time Analysis

Delivery times are heavily right-skewed because of occasional traffic delays.

Questions:

- Which measure of central tendency would you report?
- How does skewness influence business decisions?

Skills Assessed: Skewed Distribution

22. Manufacturing Process

The weight of packaged products follows an approximately normal distribution.

Questions:

- Why is assuming normality useful?
- Which quality control decisions become easier under this assumption?

Skills Assessed: Normal Distribution

23. Website Response Time

A cloud service records millions of response times daily.

Questions:

- Why is the Central Limit Theorem useful when estimating the average response time?
- Does the original data have to follow a normal distribution?

Skills Assessed: Central Limit Theorem

24. Customer Spending

Customer spending follows a highly skewed distribution.

A manager believes averages from small samples will also be highly skewed.

Questions:

- Do you agree?
- How does increasing the sample size affect the sampling distribution?

Skills Assessed: CLT, Sampling Distribution

25. Product Weight

A packaging company wants to estimate the average product weight using random samples collected every hour.

Questions:

- Why is random sampling important?
- How does the sample size affect confidence in the estimate?

Skills Assessed: Sampling Distribution

Section 5: Confidence Intervals & Hypothesis Testing

26. Marketing Campaign

A new advertisement increases sales by 4%.

How would you determine whether the increase is statistically significant?

27. Website Redesign

Conversion rate increases from 5.1% to 5.5%.

Would you recommend deploying the new design immediately? Why or why not?

28. Customer Satisfaction

A survey estimates customer satisfaction at **82% with a 95% confidence interval of 79%–85%**.

How would you explain this interval to a business stakeholder?

29. Manufacturing

Quality engineers suspect that the average diameter of manufactured bearings has shifted.

Define the null and alternative hypotheses.

30. Healthcare

A pharmaceutical company claims its new medicine reduces recovery time.

How would you design a statistical test to evaluate the claim?

31. Product Launch

Sales increased after introducing a new feature.

How would you determine whether the increase was caused by the feature rather than random variation?

32. Loan Approval

A bank changes its credit approval policy.

Default rates increase slightly.

How would you determine whether the increase is statistically significant?

33. Mobile App

A new onboarding flow increases user retention by 2%.

How would you statistically validate the improvement before rollout?

34. Manufacturing Inspection

The historical defect rate is 2%.

Today's sample shows 3.2% defects.

Should production be stopped immediately?

Explain your reasoning.

35. Customer Support

Average customer waiting time decreased by 30 seconds after introducing a chatbot.

How would you determine whether the improvement is statistically meaningful?

Section 6: Correlation & Regression

36. Ice Cream and Umbrella Sales

Ice cream and umbrella sales both increase during the rainy season.

Does one cause the other? Explain.

37. Employee Productivity

Employees who attend more training programmes receive higher performance ratings.

Can you conclude that training causes better performance?

38. House Price Prediction

Your regression model achieves an R^2 of 0.94 on training data but only 0.68 on unseen data.

What might explain the difference?

39. Sales Forecasting

A regression model consistently underestimates sales during festive seasons.

How would you improve the model?

40. Retail Pricing

A retailer wants to understand how discounts influence sales.

Would correlation analysis alone be sufficient?

Explain.

41. Customer Churn

A feature has a strong correlation with churn but very low importance in the prediction model.

How would you explain this difference?

42. Healthcare Analytics

A regression model predicts patient recovery time.

Which assumptions would you verify before trusting the model?

Section 7: Model Evaluation & Business Analytics

43. Fraud Detection

Your fraud detection model achieves **99% accuracy**, but fraud cases represent only **0.5%** of all transactions.

Would you consider this a good model? Why?

44. Medical Diagnosis

A disease detection model has high precision but low recall.

Would you deploy it in a hospital? Explain.

45. Customer Churn Prediction

Your model performs extremely well during training but poorly on new customer data.

What statistical issue is occurring?

46. Recommendation Engine

A recommendation model increases click-through rate but decreases average order value.

Which business metrics would influence your decision?

47. AI-Assisted Recruitment

An AI model shortlists significantly fewer candidates from one demographic group.

Which statistical analyses would you perform before concluding the model is biased?

48. Dashboard Validation

Before presenting a dashboard to senior management, what statistical and data quality checks would you perform?

49. Data Quality Assessment

A dataset contains duplicate records, missing values, inconsistent formats and unusual outliers.

How would you prioritise your data validation and cleaning process before analysis?

50. Executive Decision Making

The CEO asks:

> "Can you guarantee that implementing this predictive model will improve revenue?"

How would you answer using statistical reasoning and effectively communicate uncertainty?

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